Generalized Forces (Part A)

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1. Newton's Second Law in Polar Coordinates

In Cartesian coordinates, Newton's second law takes the simple form $\vec{F} = m\vec{a}$. In curvilinear coordinates, such as polar coordinates (r, θ), the components of acceleration are more complicated. The formalism of **generalized forces (广义力)** allows us to write the correct equations of motion systematically for any choice of coordinates.

We begin with Newton's second law expressed in polar basis vectors:

$$\vec{F} = m\vec{a}. \tag{1}$$

The force is decomposed as

$$\vec{F} = F_r\,\hat{r} + F_\theta\,\hat{\theta}. \tag{2}$$

The polar basis vectors are written in the Cartesian basis as

$$\hat{r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}, \tag{3}$$

$$\hat{\theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y}. \tag{4}$$

The position vector is

$$\vec{r} = r\,\hat{r}. \tag{5}$$

1.1 Generalized Forces

The **generalized force** associated with a coordinate q_i is defined by

$$Q_i = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_i}. \tag{6}$$

For the radial coordinate $q_i = r$, we have

$$\frac{\partial \vec{r}}{\partial r} = \hat{r}, \tag{7}$$

so

$$Q_r = \vec{F} \cdot \hat{r} = F_r. \tag{8}$$

For the angular coordinate $q_i = \theta$, we use

$$\frac{\partial \vec{r}}{\partial \theta} = r\frac{\partial \hat{r}}{\partial \theta}. \tag{9}$$

Since

$$\frac{\partial \hat{r}}{\partial \theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y} = \hat{\theta}, \tag{10}$$

we obtain

$$Q_\theta = \vec{F} \cdot (r\hat{\theta}) = rF_\theta. \tag{11}$$

> **Physical interpretation.** Q_r is the ordinary radial force F_r, whereas Q_θ is a torque (with units of force times length). This is natural: the generalized force conjugate to an angle is a torque, just as the generalized force conjugate to a length is an ordinary force.

1.2 Derivation from Lagrange's Equation

We now show that Lagrange's equations reproduce Newton's second law in polar coordinates. The kinetic energy is

$$T = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right). \tag{12}$$

Since there is no potential energy in this example, the Lagrangian is simply $L = T$. The Lagrange equation with a generalized force Q_i reads

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i. \tag{13}$$

1.2.1 Radial Coordinate

Setting $q_i = r$, we find

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 = Q_r = F_r. \tag{14}$$

Therefore,

$$F_r = m\ddot{r} - mr\dot{\theta}^2 = m\left(\ddot{r} - r\dot{\theta}^2 \right). \tag{15}$$

This is precisely the radial component of $\vec{F} = m\vec{a}$ in polar coordinates.

1.2.2 Angular Coordinate

Setting $q_i = \theta$, we find

$$\frac{d}{dt}\left(mr^2\dot{\theta} \right) - 0 = Q_\theta = rF_\theta. \tag{16}$$

Expanding the time derivative gives

$$2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} = rF_\theta. \tag{17}$$

Dividing by r and rearranging,

$$F_\theta = m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta} \right). \tag{18}$$

This is the angular component of $\vec{F} = m\vec{a}$.

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2. Velocity-Dependent Forces

A velocity-dependent force has the form

$$\vec{F} = \vec{F}(\vec{r}, \vec{v}). \tag{19}$$

Common examples include the magnetic Lorentz force and friction. Because such a force cannot be expressed as the gradient of an ordinary potential $V(\vec{r})$, two methods are used to incorporate it into the Lagrangian framework.

2.1 Method 1: Generalized Force

We simply define the generalized force in the usual way,

$$Q_i = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_i}, \tag{20}$$

and place it directly on the right-hand side of Lagrange's equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i. \tag{21}$$

In this approach, the dynamics are not fully encoded in the Lagrangian alone.

2.2 Method 2: Velocity-Dependent Potential

Alternatively, one may seek a **velocity-dependent potential (与速度相关的势)** $U(\vec{r}, \vec{v})$ such that the Lagrangian

$$L = T - U \tag{22}$$

contains all the dynamics. The resulting generalized force then appears automatically from the left-hand side of Lagrange's equation. However, finding such a potential for an arbitrary velocity-dependent force is not always possible.

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3. Lorentz Force

The Lorentz force on a particle of charge q moving in electric and magnetic fields is

$$\vec{F} = q\vec{E} + q\,\vec{v} \times \vec{B}. \tag{23}$$

3.1 Lorentz Force as a Generalized Force

Using Method 1, the generalized force is

$$Q_i = \vec{F} \cdot \frac{\partial \vec{r}}{\partial q_i}. \tag{24}$$

Therefore,

$$Q_i = q\vec{E} \cdot \frac{\partial \vec{r}}{\partial q_i} + q(\vec{v} \times \vec{B}) \cdot \frac{\partial \vec{r}}{\partial q_i}. \tag{25}$$

An important identity connects the derivative of the position vector with the derivative of the velocity:

$$\frac{\partial \vec{r}}{\partial q_i} = \frac{\partial \dot{\vec{r}}}{\partial \dot{q}_i} = \frac{\partial \vec{v}}{\partial \dot{q}_i}. \tag{26}$$

Hence we may write

$$Q_i = \left[q\vec{E} + q(\vec{v} \times \vec{B})\right] \cdot \frac{\partial \vec{v}}{\partial \dot{q}_i}. \tag{27}$$

3.2 Uniform Magnetic Field in the z Direction

Consider the special case

$$\vec{B} = B\,\hat{z}, \qquad \vec{E} = 0. \tag{28}$$

Then

$$Q_i = q(\vec{v} \times \vec{B}) \cdot \frac{\partial \vec{v}}{\partial \dot{q}_i}. \tag{29}$$

Evaluating the cross product,

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dot{x} & \dot{y} & \dot{z} \\ 0 & 0 & B \end{vmatrix} = B\dot{y}\,\hat{x} - B\dot{x}\,\hat{y}. \tag{30}$$

The generalized forces in Cartesian coordinates are therefore

$$Q_x = qB\dot{y}, \qquad Q_y = -qB\dot{x}, \qquad Q_z = 0. \tag{31}$$

Using the free-particle Lagrangian

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right), \tag{32}$$

Lagrange's equations give

$$m\ddot{x} = qB\dot{y}, \qquad m\ddot{y} = -qB\dot{x}, \qquad m\ddot{z} = 0. \tag{33}$$

Thus,

$$\ddot{x} = \frac{qB}{m}\dot{y}, \qquad \ddot{y} = -\frac{qB}{m}\dot{x}, \qquad \ddot{z} = 0. \tag{34}$$

> **Note.** The magnetic force acts perpendicular to the velocity, so it does no work and does not change the speed of the particle. Only the direction of motion is altered.

3.3 Helical Trajectory

A solution to the equations above is uniform motion along z combined with circular motion in the xy-plane:

$$x(t) = r\cos\omega t, \qquad y(t) = -r\sin\omega t, \qquad z(t) = v_z t, \tag{35}$$

with the **cyclotron frequency (回旋频率)**

$$\omega = \frac{qB}{m}. \tag{36}$$

Substituting this ansatz verifies that it satisfies the equations of motion. The resulting path is a **helix (螺旋线)** whose axis lies along the direction of the magnetic field.

3.4 Velocity-Dependent Potential for Electromagnetism

For a general velocity-dependent potential $U(\vec{r}, \vec{v})$, Lagrange's equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0 \tag{37}$$

yields the generalized force

$$Q_i = -\frac{\partial U}{\partial q_i} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_i}\right). \tag{38}$$

Remarkably, the Lorentz force can be derived from the potential

$$U(\vec{r}, \vec{v}) = q\phi(\vec{r}) - q\,\vec{A}(\vec{r}) \cdot \vec{v}, \tag{39}$$

where ϕ is the scalar potential and $\vec{A}$ is the vector potential, with $\vec{E} = -\nabla\phi - \partial\vec{A}/\partial t$ and $\vec{B} = \nabla \times \vec{A}$. Substituting this U into the expression for Q_i reproduces exactly the Lorentz force. Thus the full electromagnetic dynamics can be encoded in the Lagrangian $L = T - U$.

3.4.1 Detailed Derivation

For electromagnetism, we use the velocity-dependent potential

$$U(\vec{r}, \vec{v}, t) = q\phi(\vec{r}, t) - q\,\vec{A}(\vec{r}, t) \cdot \vec{v}. \tag{40}$$

In Cartesian coordinates, with repeated indices summed,

$$U = q\phi - qA_j\dot{x}_j. \tag{41}$$

Then

$$-\frac{\partial U}{\partial x_i} = -q\frac{\partial \phi}{\partial x_i} + q\frac{\partial A_j}{\partial x_i}\dot{x}_j. \tag{42}$$

Also,

$$\frac{\partial U}{\partial \dot{x}_i} = -qA_i, \tag{43}$$

so

$$\frac{d}{dt}\left(\frac{\partial U}{\partial \dot{x}_i}\right) = -q\frac{dA_i}{dt}. \tag{44}$$

Since $A_i = A_i(\vec{r}, t)$,

$$\frac{dA_i}{dt} = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial x_j}\dot{x}_j. \tag{45}$$

Therefore,

$$\frac{d}{dt}\left(\frac{\partial U}{\partial \dot{x}_i}\right) = -q\frac{\partial A_i}{\partial t} - q\frac{\partial A_i}{\partial x_j}\dot{x}_j. \tag{46}$$

Combining terms,

$$Q_i = -q\frac{\partial \phi}{\partial x_i} + q\frac{\partial A_j}{\partial x_i}\dot{x}_j - q\frac{\partial A_i}{\partial t} - q\frac{\partial A_i}{\partial x_j}\dot{x}_j. \tag{47}$$

Thus,

$$Q_i = q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t}\right) + q\left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}\right)\dot{x}_j. \tag{48}$$

Using

$$E_i = -\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} \tag{49}$$

and

$$(\vec{v} \times \vec{B})_i = \left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}\right) v_j, \tag{50}$$

with

$$\vec{B} = \nabla \times \vec{A}, \tag{51}$$

we obtain

$$Q_i = qE_i + q(\vec{v} \times \vec{B})_i. \tag{52}$$

Therefore,

$$\vec{Q} = q\left(\vec{E} + \vec{v} \times \vec{B}\right). \tag{53}$$

So the electromagnetic velocity-dependent potential gives exactly the Lorentz force:

$$\boxed{\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)}. \tag{54}$$

The corresponding nonrelativistic Lagrangian is

$$L = T - U, \tag{55}$$

so

$$L = \frac{1}{2}mv^2 - \left(q\phi - q\vec{A} \cdot \vec{v}\right), \tag{56}$$

or

$$\boxed{L = \frac{1}{2}mv^2 + q\vec{A} \cdot \vec{v} - q\phi}. \tag{57}$$

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4. Friction and Dissipation

Friction is a **non-conservative force (非保守力)**: the total mechanical energy of a macroscopic object subject to friction is not conserved. Microscopically, the energy is transferred to the innumerable degrees of freedom of the surrounding medium (heat and sound). If one could account for every atom, total energy would still be conserved; friction is a phenomenological device that lets us avoid describing all those microscopic variables explicitly.

4.1 Examples of Friction

Kinetic friction between solid surfaces is often modeled as

$$f_k = \mu N, \tag{58}$$

where N is the normal force and μ is the coefficient of friction.

A drag force in a fluid is typically velocity-dependent:

$$f_d = bv^n. \tag{59}$$

The friction force is opposite to the direction of motion:

$$\vec{f} = -bv^n\,\hat{v}. \tag{60}$$

More generally, the coefficient b may depend on time and position:

$$\vec{f} = -b(t, q_i)\,v^n\,\hat{v}. \tag{61}$$

> **Physical values.** Linear drag ($n = 1$) describes Stokes flow at low Reynolds number, while quadratic drag ($n = 2$) is appropriate for turbulent flow at high Reynolds number.

4.2 Free Particle Under Friction

Consider a free particle subject to a drag force $\vec{f} = -bv^n\hat{v}$. Its mechanical energy is purely kinetic,

$$E = T = \frac{1}{2}m\vec{v}\cdot\vec{v}. \tag{62}$$

The rate of change of energy is

$$\frac{dE}{dt} = \vec{v}\cdot m\vec{a} = \vec{v}\cdot\vec{f} = -bv^{n+1}. \tag{63}$$

Thus the energy decreases monotonically, as expected for a dissipative force.

4.2.1 Linear Drag

A simple way to understand why linear drag occurs is to consider an object moving **slowly** through a viscous fluid.

At very low speeds, the flow around the object is smooth and steady. In this regime the dominant resistance comes from the fluid's **viscosity**, not from turbulence or wake formation.

Viscosity produces a shear stress proportional to the velocity gradient,

$$\text{viscous stress} \propto \eta\frac{v}{L}, \tag{64}$$

where η is the fluid viscosity, v is the speed of the object, and L is a typical length scale, such as the radius or diameter of the object.

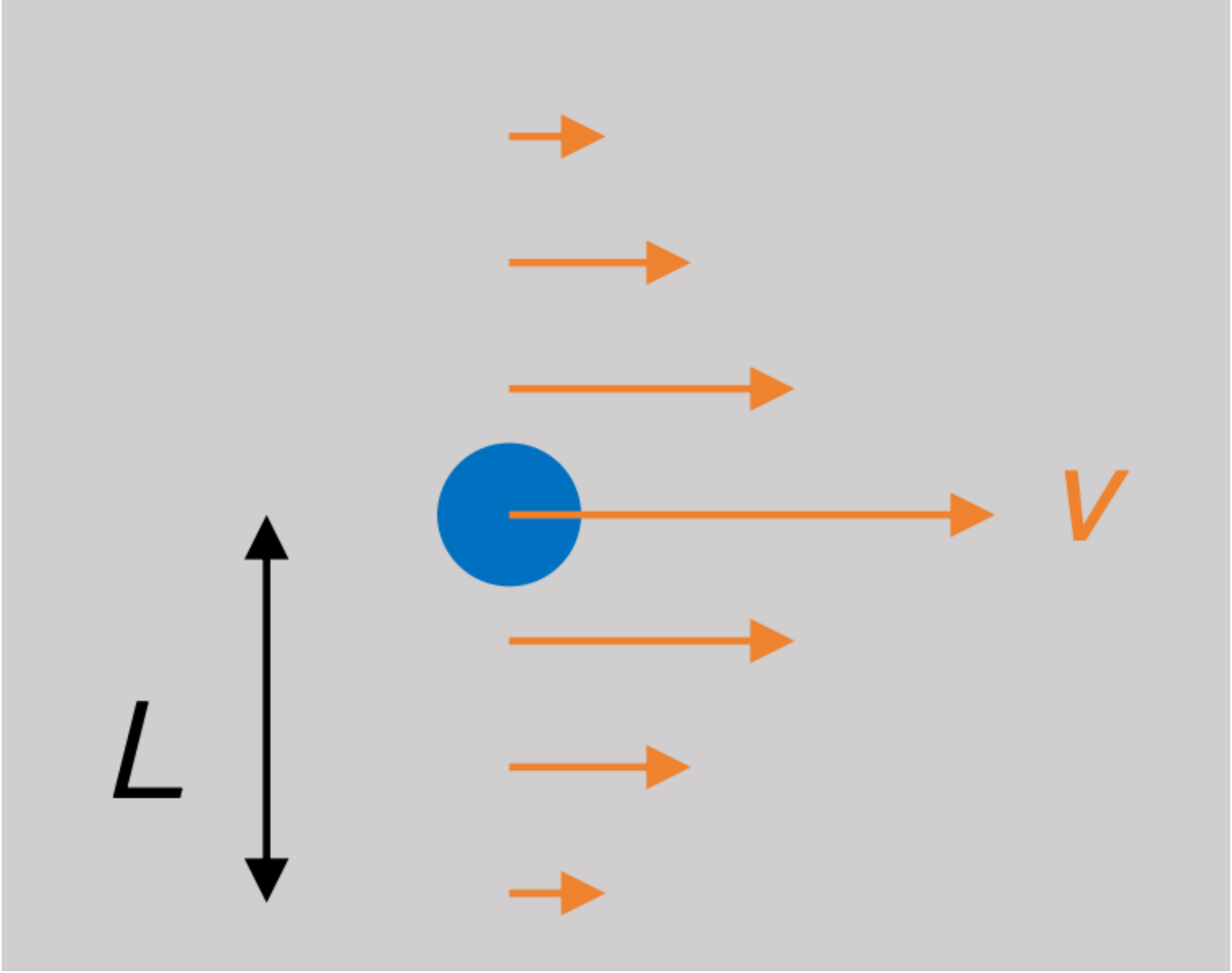


Figure 1. Viscous shear in the fluid around a slowly moving object.

The drag force is approximately the viscous stress times the area over which it acts. Since the surface area scales like L^2, we have

$$f_d \propto \left(\eta \frac{v}{L}\right) L^2. \tag{65}$$

Simplifying gives

$$f_d \propto \eta L v. \tag{66}$$

For a fixed object moving through a fixed fluid, both η and L are constants. Therefore all constant factors can be absorbed into a single coefficient b, giving

$$f_d = bv, \tag{67}$$

that is, $n = 1$.

Thus, **linear drag** arises when viscous forces dominate the motion. This occurs for slow motion through a fluid, or, more generally, when the Reynolds number is small.

4.2.2 Quadratic Drag

A simple way to understand why quadratic drag occurs is to consider an object moving **quickly** through a fluid.

At high speeds the object must continuously push fluid out of the way. In this regime the dominant resistance comes from transferring momentum to the surrounding fluid, rather than from ordinary viscous shear alone.

Suppose the object has cross-sectional area A and moves with speed v. In a time interval Δt, it sweeps out a volume of fluid approximately

$$\Delta V \sim Av\,\Delta t. \tag{68}$$

If the fluid density is ρ, the mass of fluid disturbed during this time is

$$\Delta m \sim \rho\,\Delta V \sim \rho A v\,\Delta t. \tag{69}$$

The object changes the velocity of this disturbed fluid by an amount of order v. Therefore the momentum transferred to the fluid is roughly

$$\Delta p \sim \Delta m\, v \sim \rho A v^2 \Delta t. \tag{70}$$

The drag force is the rate of momentum transfer,

$$f_d \sim \frac{\Delta p}{\Delta t} \sim \rho A v^2. \tag{71}$$

This has the same form as

$$f_d = bv^2, \tag{72}$$

where the constant b contains factors such as the fluid density, cross-sectional area, and shape of the object. More precisely, quadratic drag is often written as

$$f_d = \frac{1}{2}\rho C_D A v^2, \tag{73}$$

where C_D is the drag coefficient. Thus $n = 2$.

Therefore, **quadratic drag** arises when inertial effects dominate. Physically, this means the object loses energy mainly by pushing fluid aside and creating a wake (尾流).

> **Definition.** A wake is the downstream flow disturbance left behind a body because of momentum loss, flow separation, and viscous effects.

4.3 Generalized Friction Force and the Dissipation Function

The generalized friction force is defined by projecting the friction force onto the generalized coordinate:

$$f_{n,i} = \vec{f} \cdot \frac{\partial \vec{r}}{\partial q_i}. \tag{74}$$

Using the identity $\partial\vec{r}/\partial q_i = \partial\vec{v}/\partial\dot{q}_i$ and $\hat{v} = \vec{v}/v$, we find

$$f_{n,i} = -bv^n\,\hat{v}\cdot\frac{\partial\vec{v}}{\partial\dot{q}_i} = -bv^n\frac{\partial v}{\partial\dot{q}_i}. \tag{75}$$

Because

$$\frac{\partial}{\partial\dot{q}_i}\left(\frac{1}{n+1}bv^{n+1}\right) = bv^n\frac{\partial v}{\partial\dot{q}_i}, \tag{76}$$

the generalized friction force can be written as

$$f_{n,i} = -\frac{\partial D_n}{\partial\dot{q}_i}, \tag{77}$$

where the **dissipation function (耗散函数)** is

$$D_n(v) = \frac{1}{n+1}bv^{n+1}. \tag{78}$$

Lagrange's equation including friction then takes the form

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_i}\right) - \frac{\partial L}{\partial q_i} + \frac{\partial D_n}{\partial\dot{q}_i} = 0. \tag{79}$$

The dissipation function is directly related to the rate of energy loss:

$$\frac{dE}{dt} = -(n+1)D_n(v). \tag{80}$$

Hence D_n quantifies the power dissipated from the macroscopic motion.

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5. Damped Pendulum

Consider a simple pendulum of length l and mass m, subject to linear damping

$$f_d = bv. \tag{81}$$

The particle speed is $v = l\dot{\theta}$, and the Lagrangian is

$$L = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta. \tag{82}$$

> **Note on the sign.** Here $\theta = 0$ is the stable equilibrium (hanging straight down). The potential energy is $V = -mgl\cos\theta$, so $L = T - V$ takes the form above.

For linear damping ($n = 1$) the dissipation function is

$$D_1 = \frac{1}{2}bv^2 = \frac{1}{2}bl^2\dot{\theta}^2. \tag{83}$$

The modified Lagrange equation reads

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} + \frac{\partial D_1}{\partial \dot{\theta}} = 0. \tag{84}$$

Evaluating each term,

$$\frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}, \qquad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = ml^2\ddot{\theta}, \qquad \frac{\partial L}{\partial \theta} = -mgl\sin\theta, \qquad \frac{\partial D_1}{\partial \dot{\theta}} = bl^2\dot{\theta}. \tag{85}$$

Collecting terms,

$$ml^2\ddot{\theta} + mgl\sin\theta + bl^2\dot{\theta} = 0. \tag{86}$$

Dividing by ml^2,

$$\ddot{\theta} = -\frac{g}{l}\sin\theta - \frac{b}{m}\dot{\theta}. \tag{87}$$

For small oscillations, $\sin\theta \simeq \theta$, and the equation linearizes to

$$\ddot{\theta} + \frac{b}{m}\dot{\theta} + \frac{g}{l}\theta \simeq 0. \tag{88}$$

In the underdamped regime $b^2 < 4m^2g/l$, the solution is

$$\boxed{\theta(t) = A\,e^{-\frac{bt}{2m}}\sin\left(\sqrt{\frac{g}{l} - \frac{b^2}{4m^2}}\;t + \phi\right)} \tag{89}$$

where A and ϕ are constants fixed by the initial conditions.